Longitudinal Study A - simulated results, frozen v1.0

- **Run:** 20261019T162847Z-longitudinal-full
- **Protocol:** laminar-vr-longitudinal-simulation-v1.0; investigator freeze date 2026-10-19
- **Source commit:** 4886f73b34c3589cf35b7b6f7f83dceacc6eca71; dirty at run: false
- **Unity:** 6000.0.82f1; visual bundle temple-pond-pilot-freeze-v3; mapping temple-pond-mapping-pilot-v4
- **Validation:** passed; complete frozen matrix true

Primary decision-quality result

The prespecified contrast is Carry minus Reset mean safe-oracle probe pseudo-regret after session 20, equally weighted across A1-A7. Negative values favor Carry. Synthetic profiles, not people, are the replication units.

- Paired mean: **-0.1445**; median -0.1606; seed-index SD 0.0694.
- 95% paired bootstrap interval: **[-0.1687, -0.1205]**; two-sided 100,000-draw sign-flip p = **9.9999e-06**.
- Directional criterion met: **true**. Carry was better for 100.0% of the 30 equally weighted seed-index contrasts and 83.3% of the 210 paired archetype profiles.
- Mean baseline probe regret: Carry 0.4680, Reset 0.4680; after session 20: Carry 0.3232, Reset 0.4678.

The Monte Carlo p-value is at the 100,000-draw add-one resolution; it is not an exact zero-probability result. The bootstrap interval describes variation across synthetic seed indexes under this simulator.

Archetype	Carry baseline	Reset baseline	Carry after 20	Reset after 20	Carry - Reset	Profiles favoring Carry
A1PreferenceAlignedBalanced	0.3763	0.3763	0.3017	0.3779	-0.0762	23/30
A2IlluminationDominant	0.4366	0.4366	0.3668	0.4149	-0.0481	22/30
A3WarmthDominant	0.6479	0.6479	0.6026	0.6586	-0.0560	24/30
A4SoftnessRichnessInteraction	0.6471	0.6471	0.5091	0.6555	-0.1464	27/30
A5MotionAverseThreshold	0.7240	0.7240	0.1994	0.7282	-0.5288	30/30
A6PreferencePhysiologyConflict	0.3735	0.3735	0.2215	0.3691	-0.1476	29/30
A7WeakResponder	0.0706	0.0706	0.0616	0.0703	-0.0087	20/30

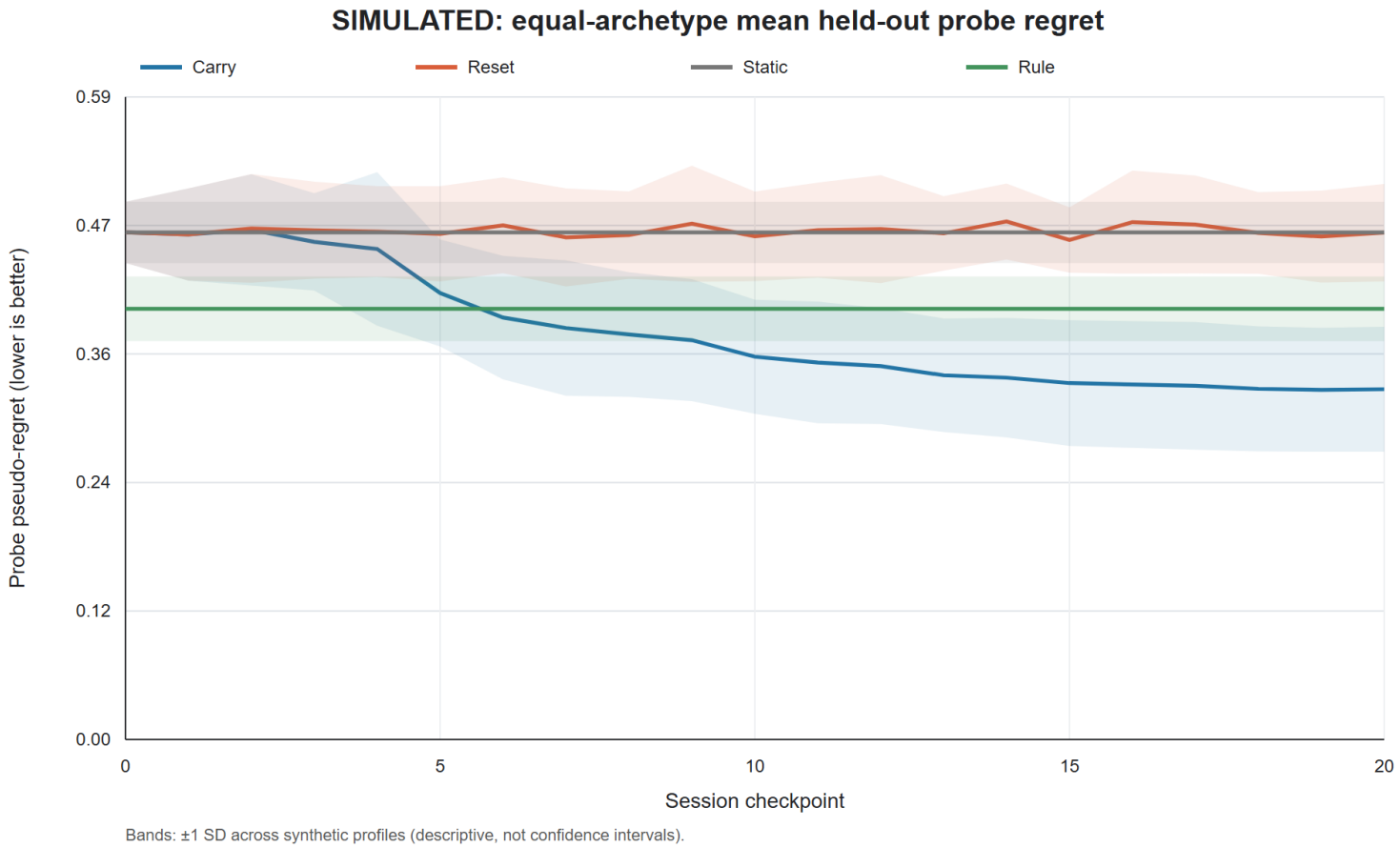


Figure 1. Simulated held-out probe pseudo-regret by session checkpoint. The pooled curve weights A1-A7 equally. Bands are +/-1 SD across synthetic seed-index aggregates, not confidence intervals.

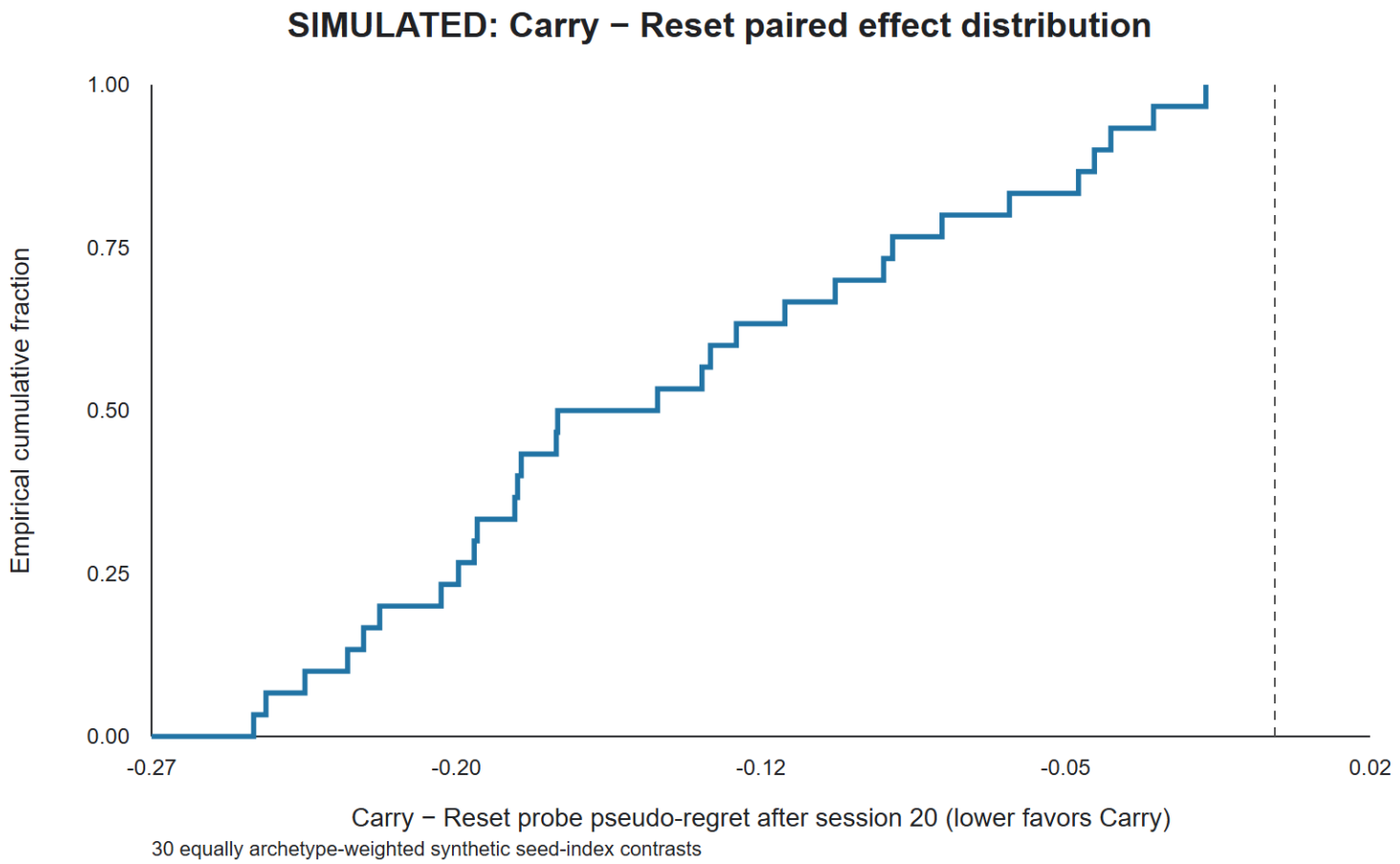


Figure 2. Distribution of the 30 equally archetype-weighted paired seed-index contrasts after session 20.

Longitudinal and challenge results (descriptive)

The seven individual A1-A7 learning curves and their between-profile SD bands are in figures/primary-01-probe-learning.svg through primary-07-probe-learning.svg. A5 has the largest mean advantage; A7 has the smallest. This heterogeneity is retained in the table rather than hidden by the pooled estimate.

Challenge	Checkpoint	Carry probe regret	Reset probe regret	Carry - Reset
BetweenSessionDrift	0	0.3709	0.3709	0.0000
BetweenSessionDrift	10	0.2916	0.3678	-0.0762
BetweenSessionDrift	11	0.3292	0.3579	-0.0287
BetweenSessionDrift	12	0.3264	0.3522	-0.0258
BetweenSessionDrift	20	0.3036	0.3531	-0.0496
MildPhysiologyLoss	0	0.3709	0.3709	0.0000
MildPhysiologyLoss	10	0.3270	0.3646	-0.0377
MildPhysiologyLoss	11	0.3333	0.3722	-0.0389
MildPhysiologyLoss	12	0.3196	0.3718	-0.0522
MildPhysiologyLoss	20	0.3266	0.3694	-0.0428
NonResponder	0	0.0000	0.0000	0.0000
NonResponder	10	0.0000	0.0000	0.0000
NonResponder	11	0.0000	0.0000	0.0000
NonResponder	12	0.0000	0.0000	0.0000
NonResponder	20	0.0000	0.0000	0.0000

For the drift challenge, the response optimum switches at session 11. A9 is a negative control with zero modeled environment effect, so its safe-oracle probe regret is zero for both policies by construction. Neither challenge is an additional confirmatory test.

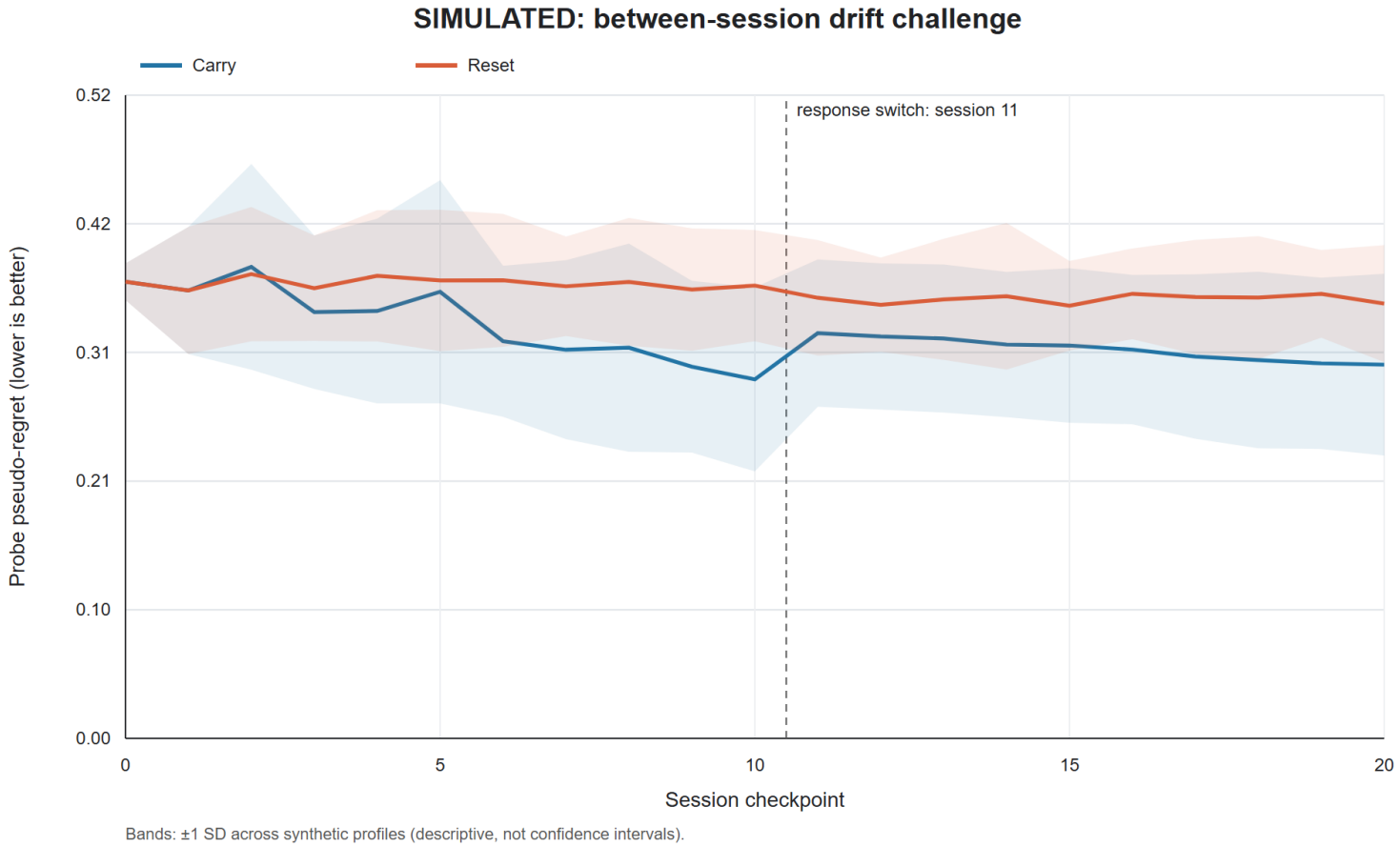


Figure 3. Simulated drift challenge with the session-11 response switch marked; bands show +/-1 SD across synthetic profiles.

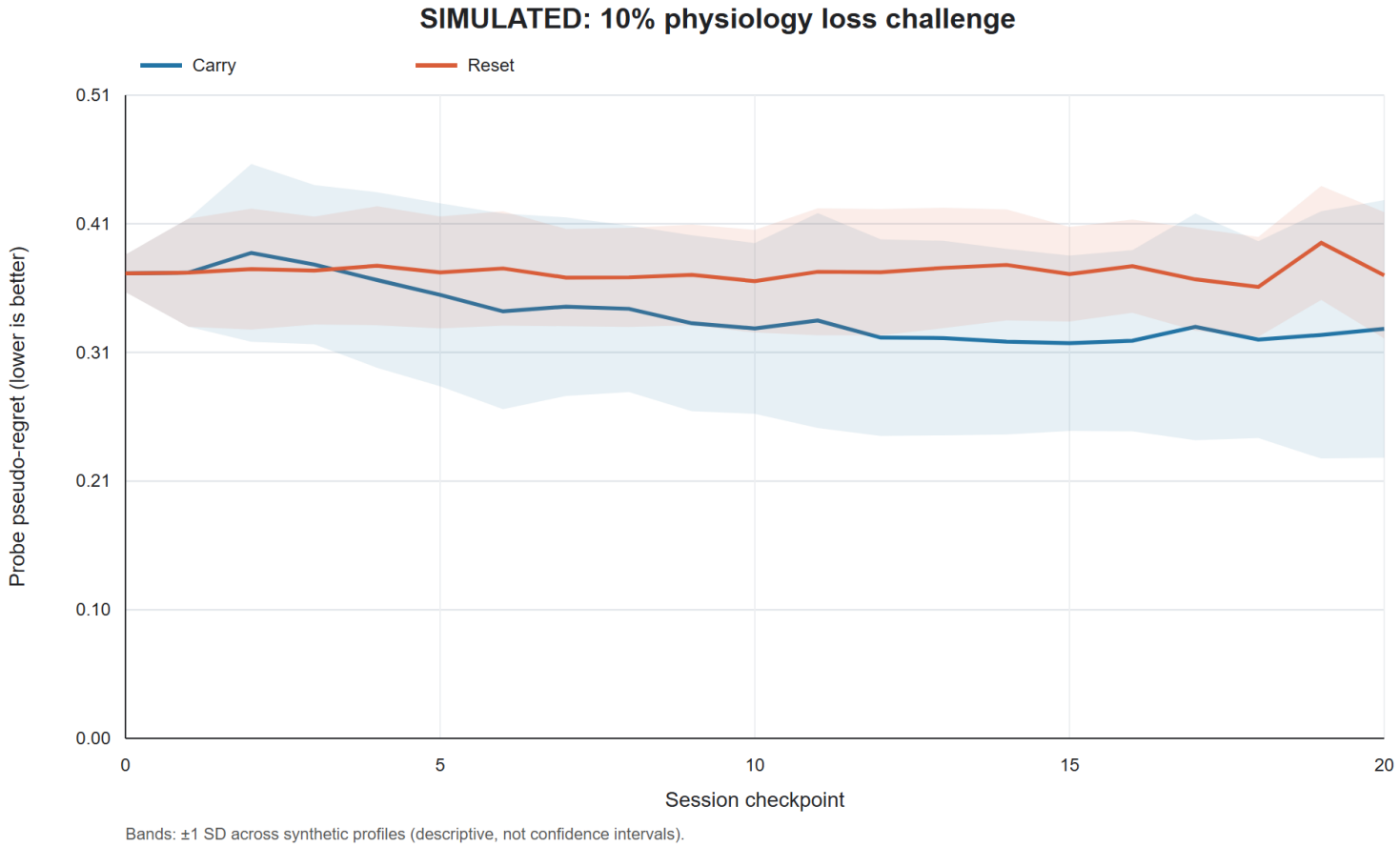


Figure 4. Simulated 10% physiology-emission loss challenge; bands show +/-1 SD across synthetic profiles.

The full primary policy comparison by early (sessions 1-5), middle (6-15), and late (16-20) periods follows. Trajectory regret uses eligible decisions as denominator, and availability is shown separately. These are descriptive totals across synthetic profiles, not independent-session tests.

Period	Policy	Eligible decisions	Valid outcomes	Trajectory regret / eligible decision	Stress-change available sessions	Mean synthetic stress change / available session
early	Carry	8400	8400	0.2678	1050	-0.5139
early	Reset	8400	8400	0.2847	1050	-0.5013

Period	Policy	Eligible decisions	Valid outcomes	Trajectory regret / eligible decision	Stress-change available sessions	Mean synthetic stress change / available session
early	Static	8400	8400	0.3003	1050	-0.6799
early	Rule	9019	6406	0.1575	1050	-0.7146
middle	Carry	16800	16800	0.1135	2100	-0.8136
middle	Reset	16800	16800	0.2791	2100	-0.4722
middle	Static	16800	16800	0.3003	2100	-0.6813
middle	Rule	18148	12528	0.1532	2100	-0.7161
late	Carry	8400	8400	0.0568	1050	-0.9417
late	Reset	8400	8400	0.2802	1050	-0.4679
late	Static	8400	8400	0.3003	1050	-0.6812
late	Rule	9028	6350	0.1593	1050	-0.7218

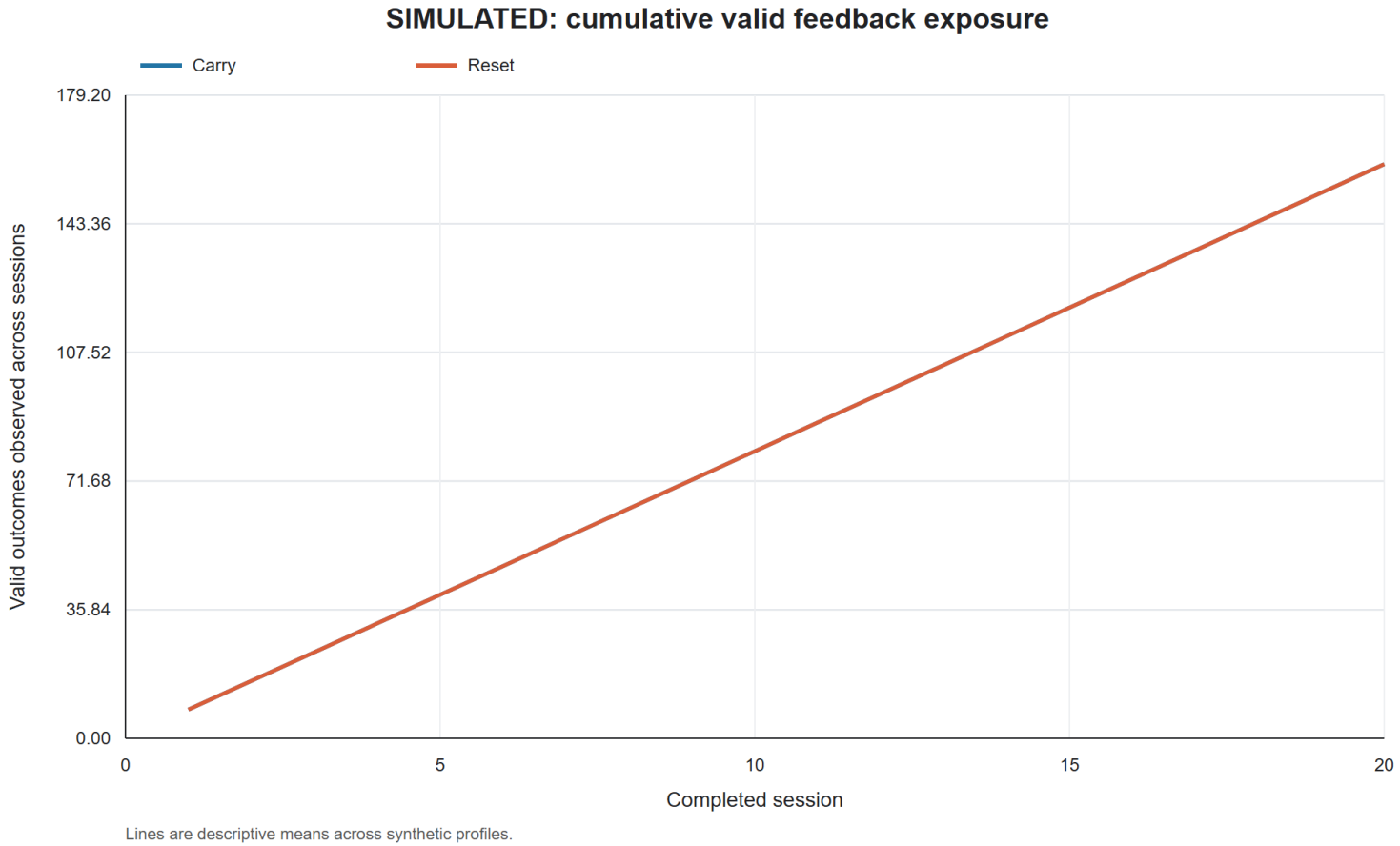


Figure 5. Mean cumulative valid feedback across primary synthetic profiles. Reset observes feedback but starts each session with a fresh LinUCB model.

Safety, attribution, and availability

Across 20,400 simulated sessions, the recorded scene-limit violations, multi-dimension violations, and invalid model updates were 0, 0, and 0. The safety layer rejected 10,911 proposals, while 33,941 visual transitions started. Rejections are expected safety actions, not executed violations.

Recorded reward events: 152,774 applied, 340 invalidated. There were 188 zero-eligible sessions, 188 baseline-unavailable sessions, and 2,230 dropped synthetic physiology emissions. Complete counts by scenario and condition are in safety-summary.csv.

Accepted-window synthetic stress, heart-rate, and RMSSD changes and observed-window stress AUC are in physiology-summary.csv. The AUC uses trapezoids between accepted samples only and is not a complete continuous-physiology integral when emissions are missing.

Interpretation and reporting limits

This result supports learning of the declared synthetic response laws by evaluation-only cross-session LinUCB carry-over. It does not establish stress reduction, safety, usability, or comfort in people. The deployed pilot still resets LinUCB each session. Its feature schema has no explicit session-context features. Study B human outcomes must be reported separately.

The raw CSVs do not separately report final environment distance to the simulator optimum, snapshot-restore failure counts, reused physiology-window counts, or pause/network boundary violations. A completed trajectory and analyzer continuity check do not replace those direct measures. Do not report them as zero. This is a reporting gap against the frozen secondary-outcome checklist.

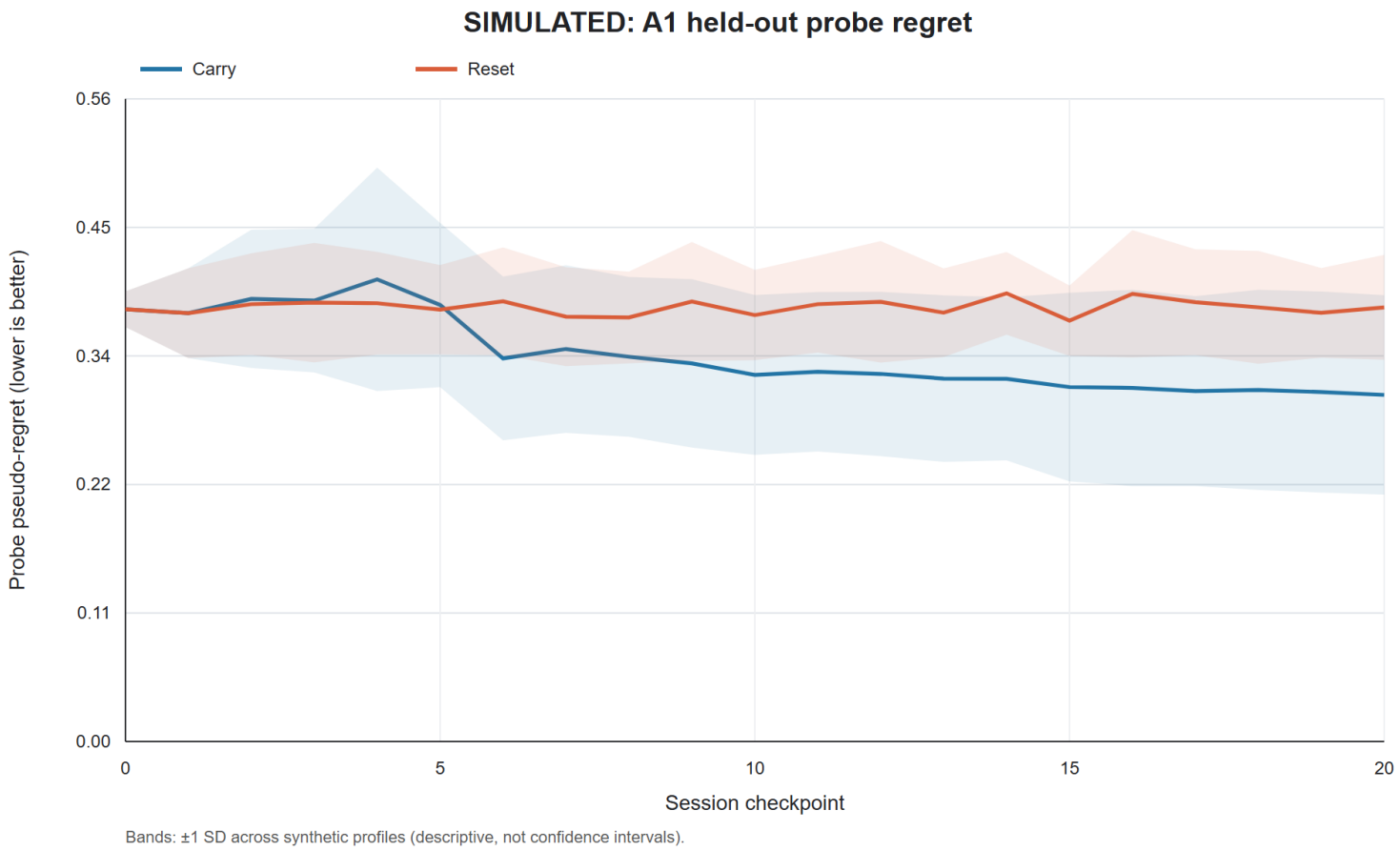
The optional three-checkpoint practical-margin endpoint was omitted before the full run. No additional inferential test was added after inspecting the full-study output. Supervisor/team sign-off remains unrecorded in the repository.

Reproducibility files

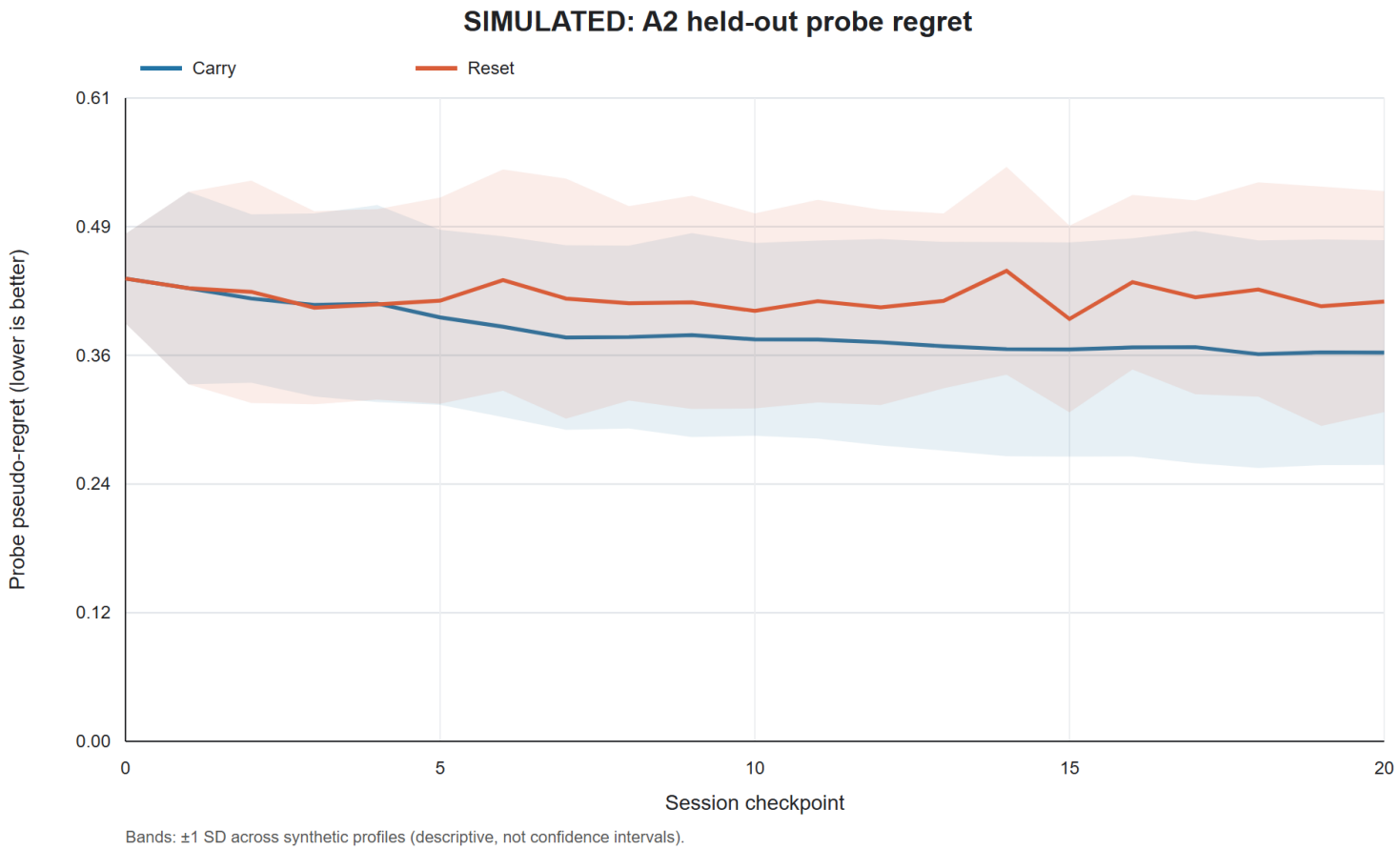
primary-archetype.csv, probe-curve.csv, session-curve.csv, period-summary.csv, safety-summary.csv, and physiology-summary.csv contain the descriptive values. report-manifest.json records the raw source and generated-file checksums. All charts are explicitly labelled simulated SVGs. The raw run folder is C:/My partition/SLIIT/Year 4/Semester 1/Research Project/R26-DS-006/analysis/component-c-bandit-evaluation/longitudinal/runs/20261019T162847Z-longitudinal-full.

Appendix: individual synthetic response curves

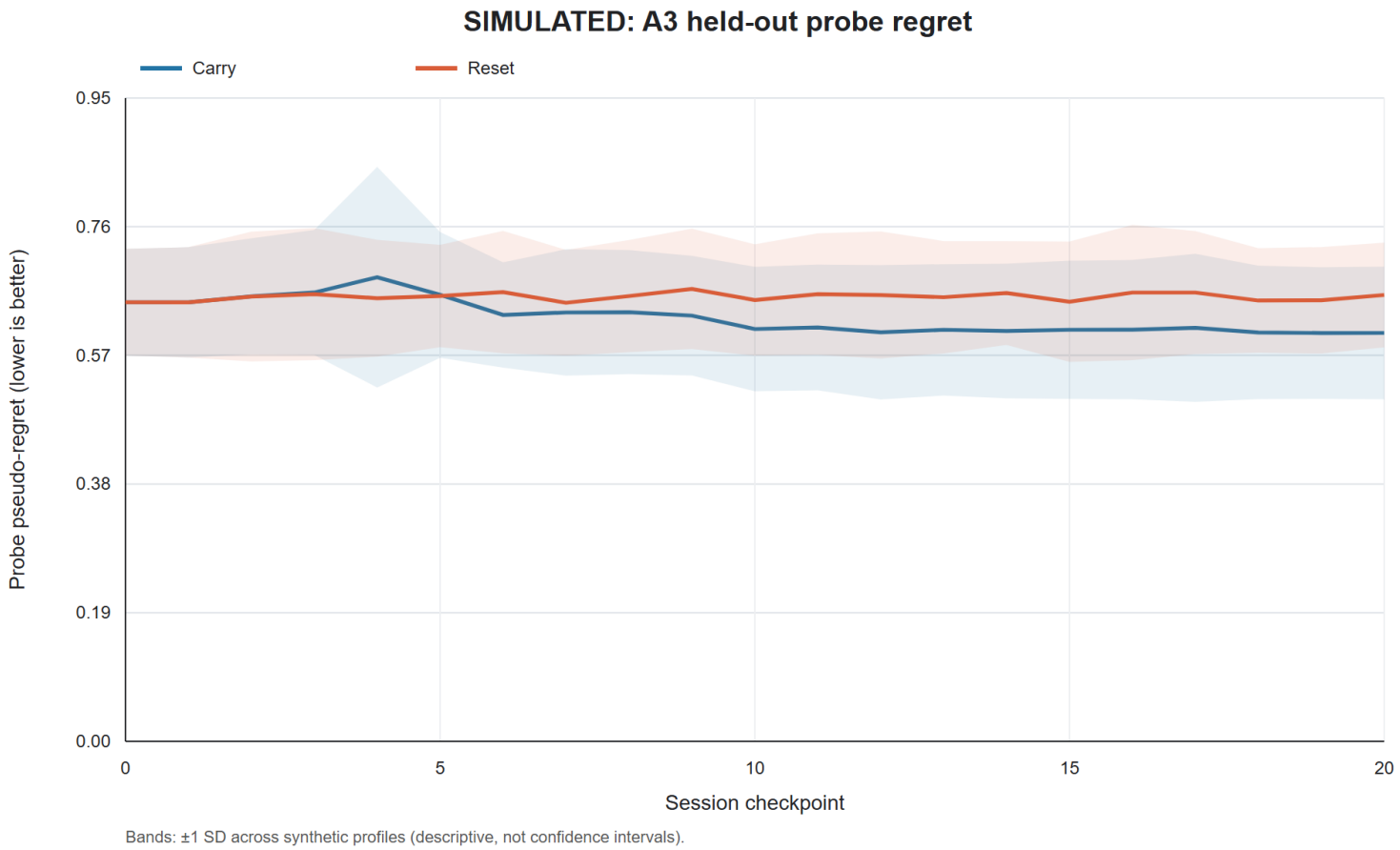
Each curve shows held-out safe-oracle probe pseudo-regret across session checkpoints for one primary archetype. Bands show between-profile standard deviation. These are descriptive simulator outputs.



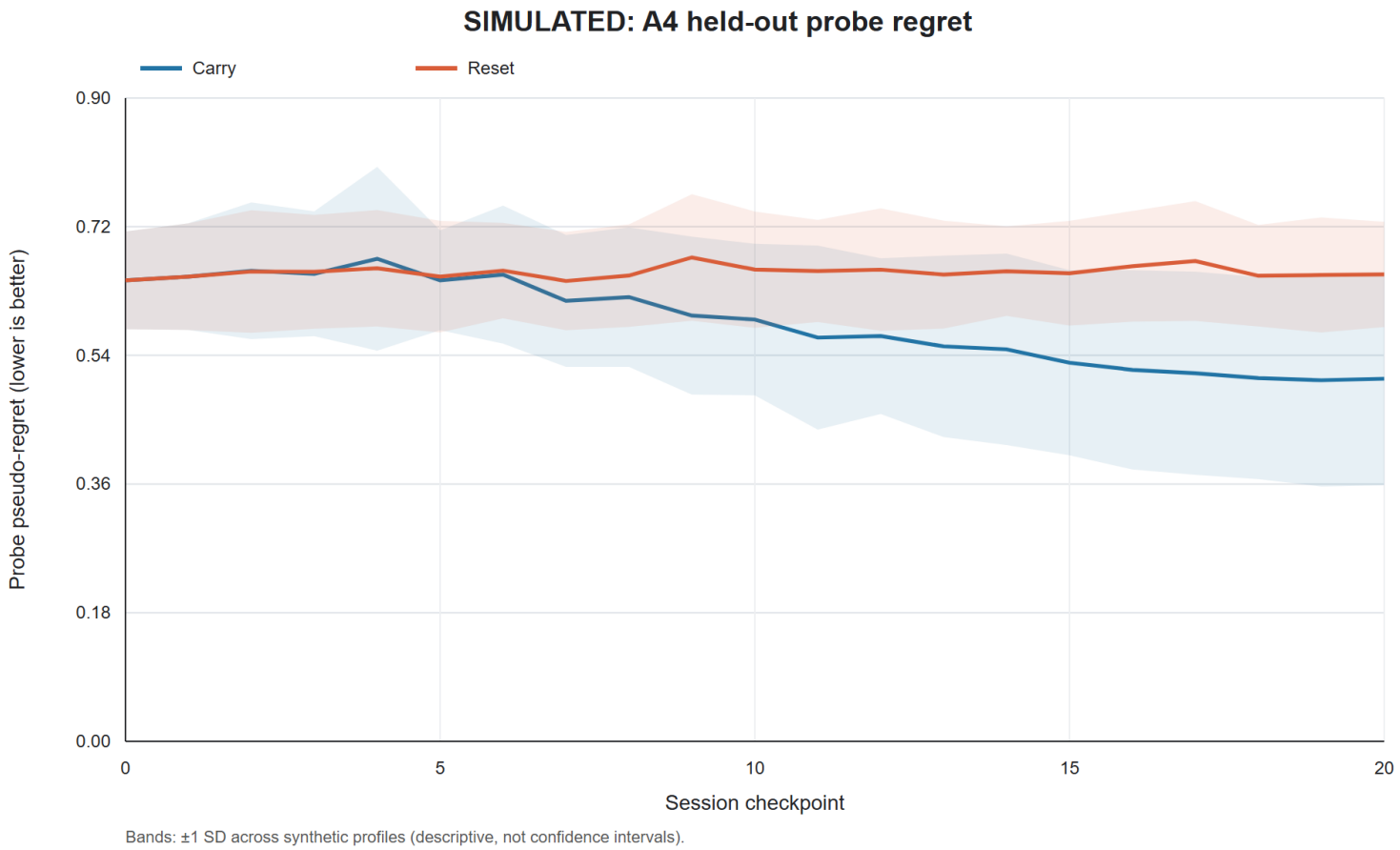
Appendix Figure A1. Simulated A1 Preference-aligned balanced held-out probe pseudo-regret.



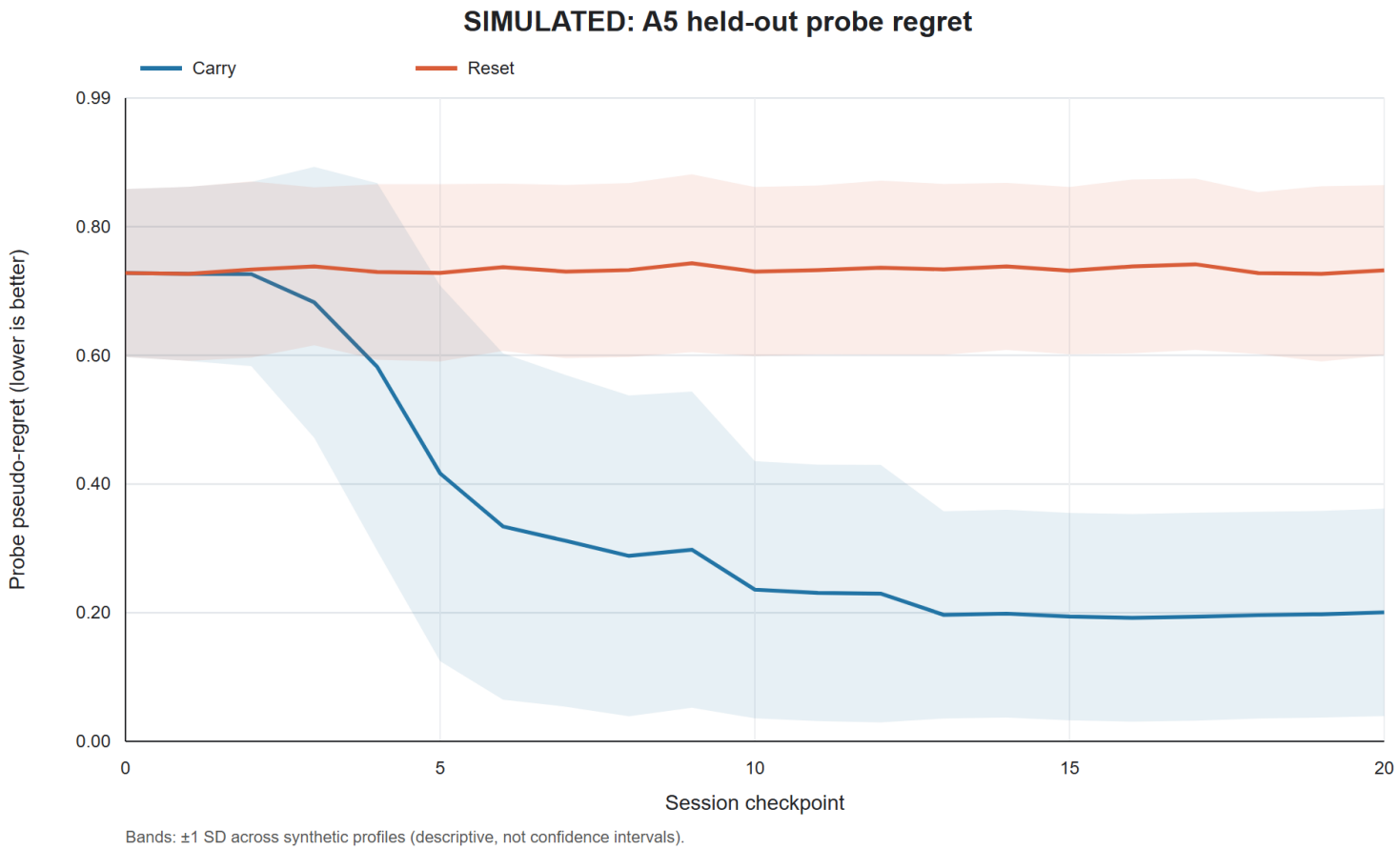
Appendix Figure A2. Simulated A2 Illumination-dominant held-out probe pseudo-regret.



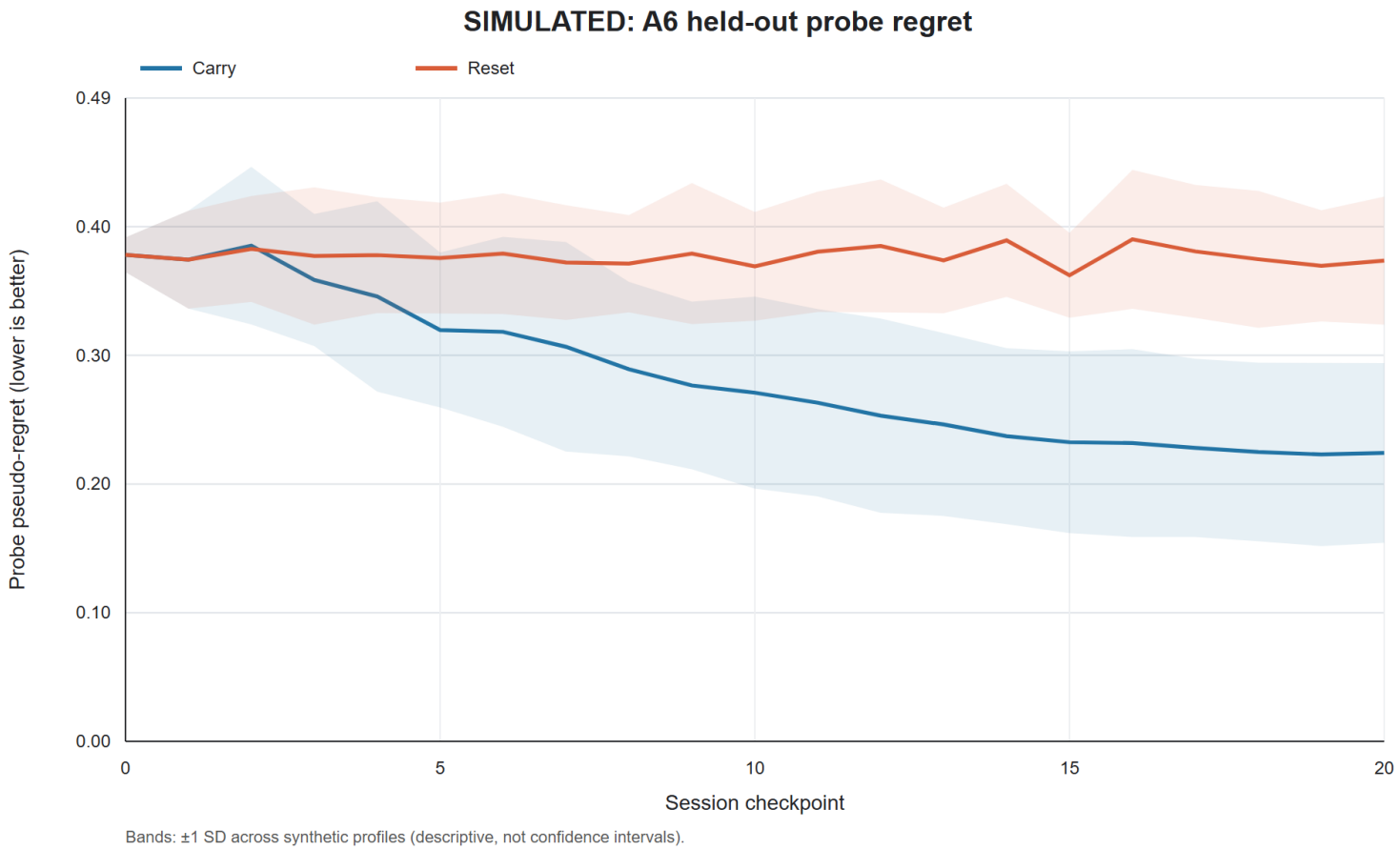
Appendix Figure A3. Simulated A3 Warmth-dominant held-out probe pseudo-regret.



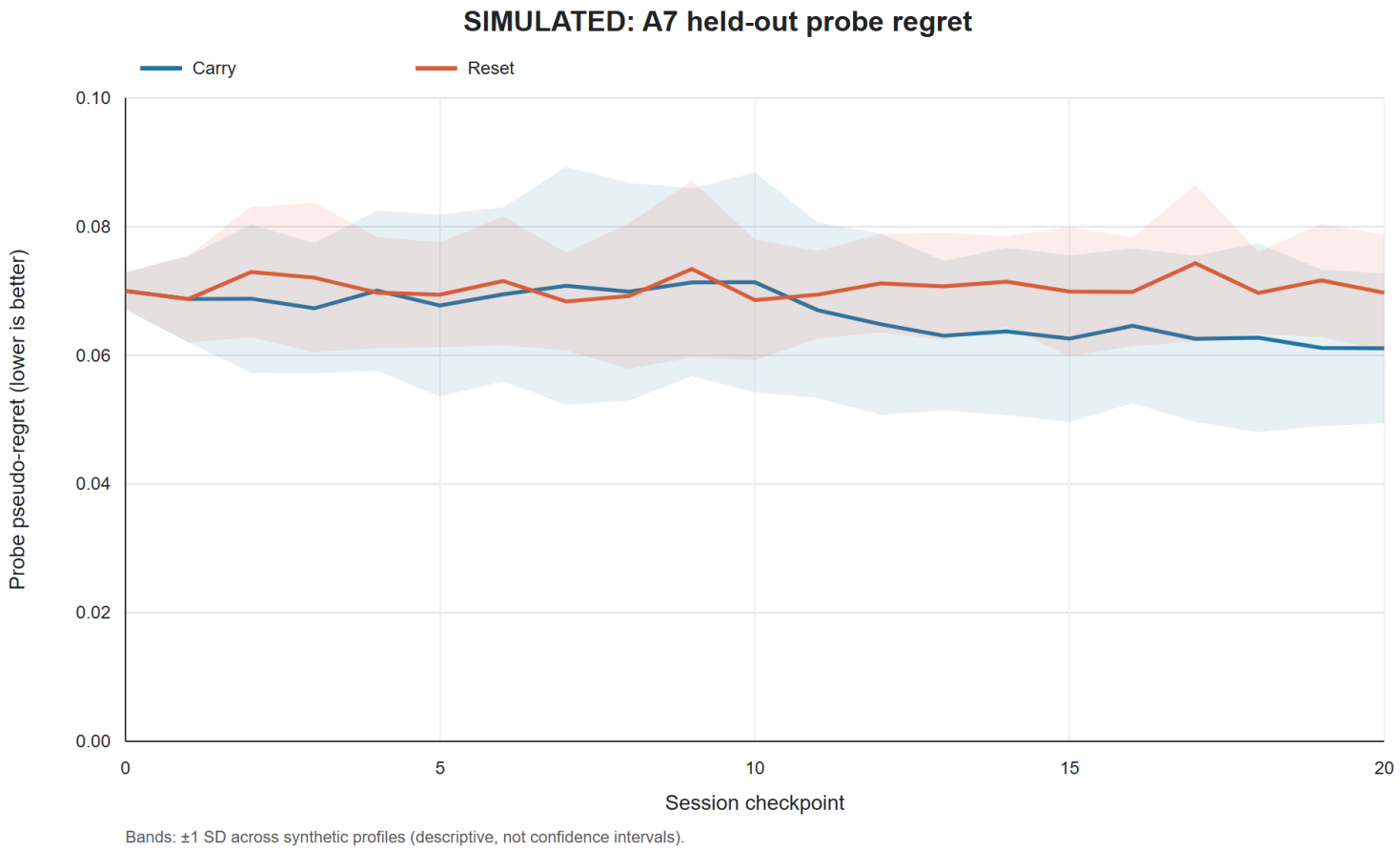
Appendix Figure A4. Simulated A4 Softness-richness interaction held-out probe pseudo-regret.



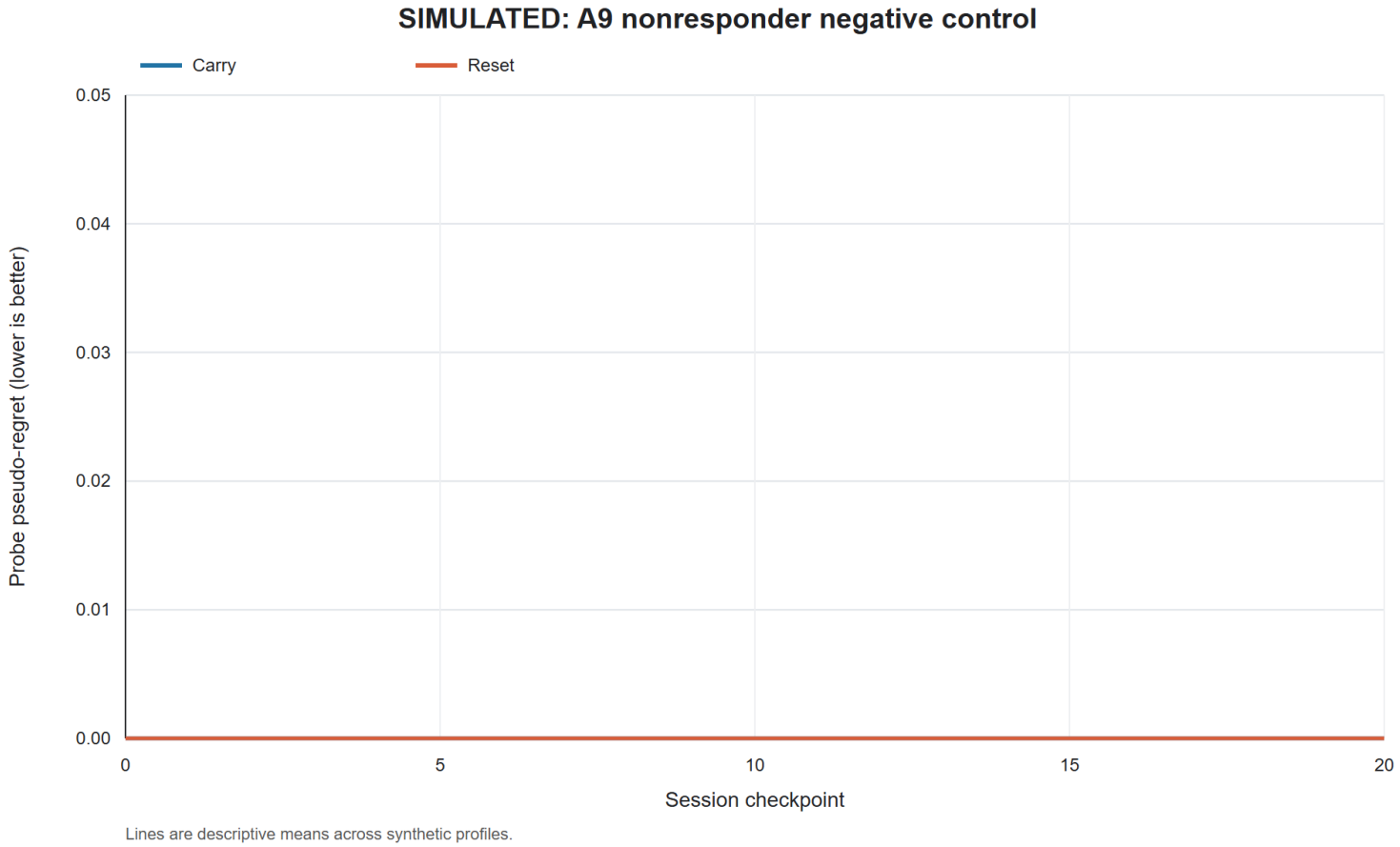
Appendix Figure A5. Simulated A5 Motion-averse threshold held-out probe pseudo-regret.



Appendix Figure A6. Simulated A6 Preference-physiology conflict held-out probe pseudo-regret.



Appendix Figure A7. Simulated A7 Weak responder held-out probe pseudo-regret.



Appendix Figure A8. Simulated non-responder negative control; both policies have zero safe-oracle probe regret by construction.